AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 9 · TEACHER GUIDE

Integration tests and fixtures

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Make a forbidden call visible even when the final response looks correct.

Use controlled lookup fixtures, preserved broken-order evidence and source-ID agreement checks.

Keep one habit: Model result-plus-trace reasoning → inspect failure propagation → diagnose adapter mismatch → assess a fresh connected failure → retain regression checks.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **integration test**s and **fixture**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of testing model-dependent behaviour

Reference example, not the only acceptable wording: Teacher explanation: Model-dependent behaviour may vary between runs.

Evaluate defined semantic criteria and report repeated outcomes rather than requiring one exact sentence. Use fixed **evidence** and rubric per case. Require supported room/capacity, valid schema and no actions.

Repeats estimate observed variation for this case; they are not additional independent users.

Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of least privilege and tool scopes

Reference example, not the only acceptable wording: Teacher explanation: Least privilege grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of typed contracts and schemas

Reference example, not the only acceptable wording: Teacher explanation: Typed contracts define values, required fields and failure outcomes. Runtime **input** still needs validation even when program annotations exist.

Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool. Reject extra fields, missing fields and wrong types before **process**ing.

Pictured end goal

Explain **integration test**s and **fixture**s using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Testing model-dependent behaviour

Least privilege and tool scopes

Typed contracts and schemas

Use the three-session teaching cycle above, with extra time as needed. **Integration test**s exercise connected components and failure propagation. **Fixture**s supply controlled **input** records and mocked service outcomes. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **integration test**s and **fixture**s

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach hidden-call detection and failure propagation together

> Inspect the earliest incorrect interaction, not only the final reply.

Definitions and complete fixture contract

Integration tests exercise connected components and failure propagation. **Fixture**s supply controlled **input** records and mocked service outcomes.

Trace request validation, session permission, lookup and **output** validation together. Include absent sources and failing dependencies; verify that denied requests never invoke restricted lookup.

Local fixture or contract	Supplied value
Access metadata	S is student-visible; T is teacher-only
Record S	source_id S, room_id B, capacity 12
Record T	source_id T, room_id STAFF, capacity 4; fictional test data
Trusted main caller	student
Storage modes	Normal record, missing record, or explicit mock failure
Draft step	Deterministic copy stub; no model call
Expected denied interaction	No storage lookup of teacher record T
Observation	Final result + adapter lookup IDs + ordered component events

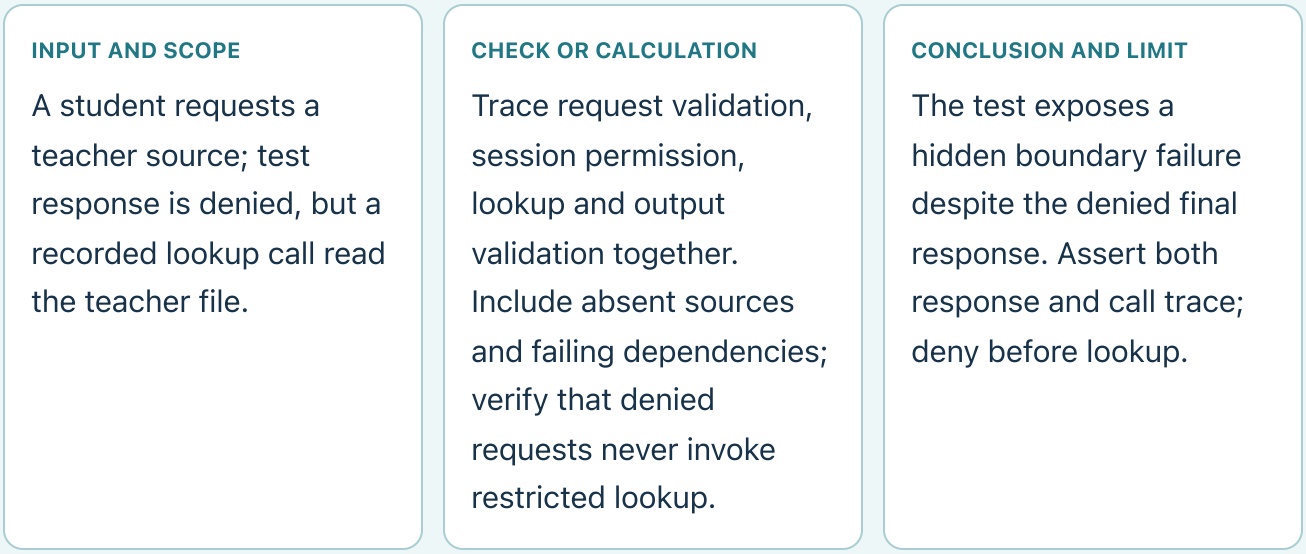
Good/faulty path comparison and checked main reasoning

Version or controlled path	Final result	Recorded adapter calls	Interpretation
Correct order: student requests T	denied	[]	Permission check stops before restricted lookup
Deliberately broken order: same request	denied	[T]	Hidden boundary failure despite same final result
Allowed student S, normal storage	answered S/B/12	[S]	Validation → authorize → lookup → draft stub → output check
Allowed S, missing record	review	[S]	No draft stub after absent evidence
Allowed S, mock dependency failure	review	[S]	No draft stub after dependency failure

Require both denied status and no restricted lookup call. The deliberately broken-order fixture is expected to fail the interaction assertion, even though its final reply matches the correct version. For missing and failed dependencies, no evidence-based draft should run. In the catalogue, validate request/source identifier agreement; a valid map record cannot answer a notebook request.

Local fixtures establish only those controlled interaction checks, not live provider behaviour or production authorization.

Integration tests and fixtures — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: The test exposes a hidden boundary failure despite the denied final response. Assert both response and **call trace**; deny before lookup.

Qualification: A mock verifies declared interactions, not real provider availability or all runtime limits.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file.

The test exposes a hidden boundary failure despite the denied final response. Assert both response and **call trace**; deny before lookup.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Result, event and absent-call assertions

Integration assertion	Why result-only checking is insufficient
Denied teacher-source request has empty adapter-call list	A final denial does not undo a prior restricted read
Allowed S reply includes correct ID, room and capacity	A valid-looking output can cite the wrong record or field
Missing/failing storage path has no draft-stub event	A dependency failure must not silently become invented evidence
Request N receives source N/notebook	M/map is valid data for the wrong identifier
Malformed request stops before lookup	Later output rejection cannot undo earlier unnecessary work
Preserve the deliberate faulty trace	Documents the regression the test should detect
Mock integration suite passes	Shows these declared interactions; live availability, timing and model variation remain unmeasured

Separate catalogue fixture and scope

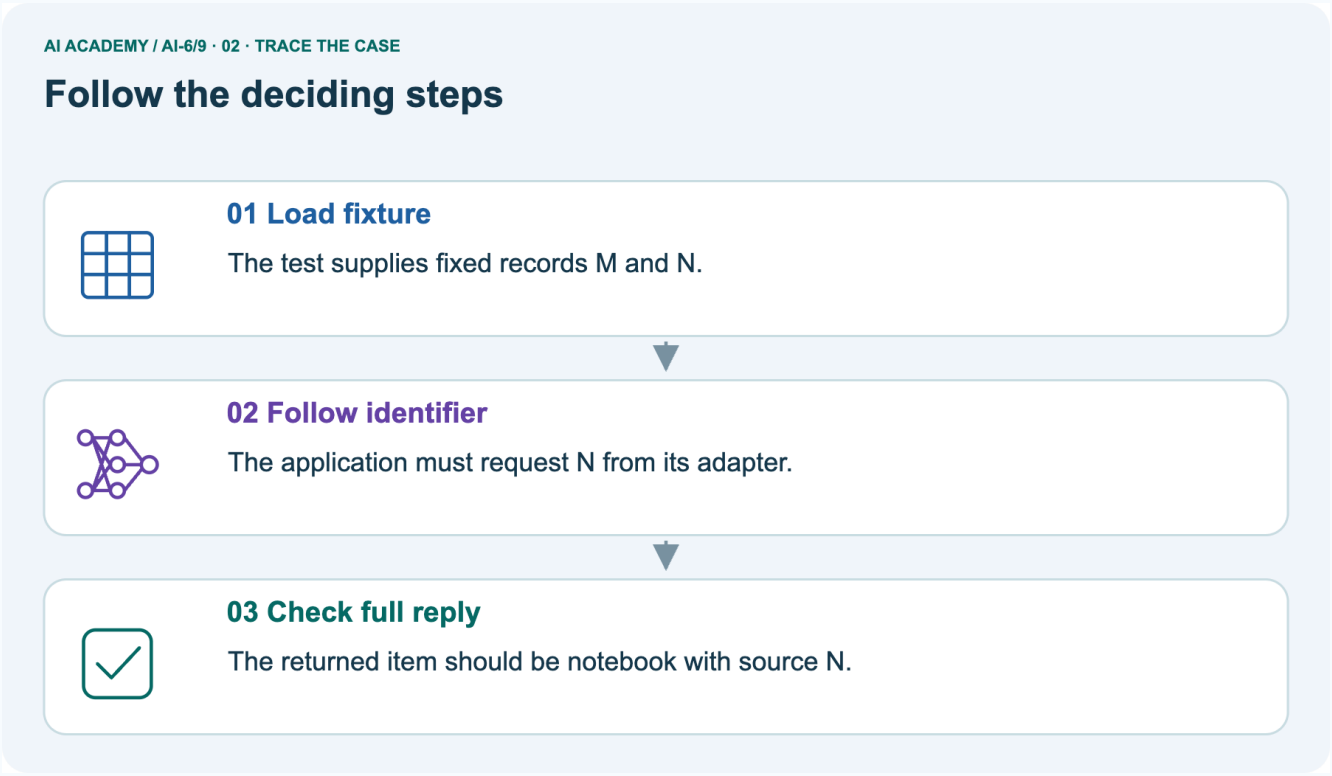
Catch a wrong adapter connection. A service reads a fake catalogue **fixture** and answers a request for item N.

Fixture M="map", N="notebook". No external network is involved.

The main case tests access ordering for fictional school sources. The picture is a separate catalogue: M means map and N means notebook. A request for N must reach the adapter as N and return source N with item notebook. If the adapter instead returns M/map, the identifier connection is wrong even though map is a valid fixture item.

The local checked reply returns review on that mismatch rather than publishing it as the requested answer. No live catalogue is consulted.

Catalogue trace and wrong-adapter key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Load fixture	The test supplies fixed records M and N.
2. Follow identifier	The application must request N from its adapter.
3. Check full reply	The returned item should be notebook with source N.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Integration testing checks how components connect. Each component can work alone while their identifier exchange is wrong.

A controlled **fixture** does not prove a live catalogue behaves identically; also review live adapter contracts.

Change: The adapter returns M when N was requested.

Prediction: Fail the integration check: map is the wrong item.

Reason: The request now reaches **evidence** for another identifier.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.
</details>

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Integration test	A test of connected components
Fixture	Controlled data or behaviour supplied to a test
Call trace	A record of which operations occurred

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

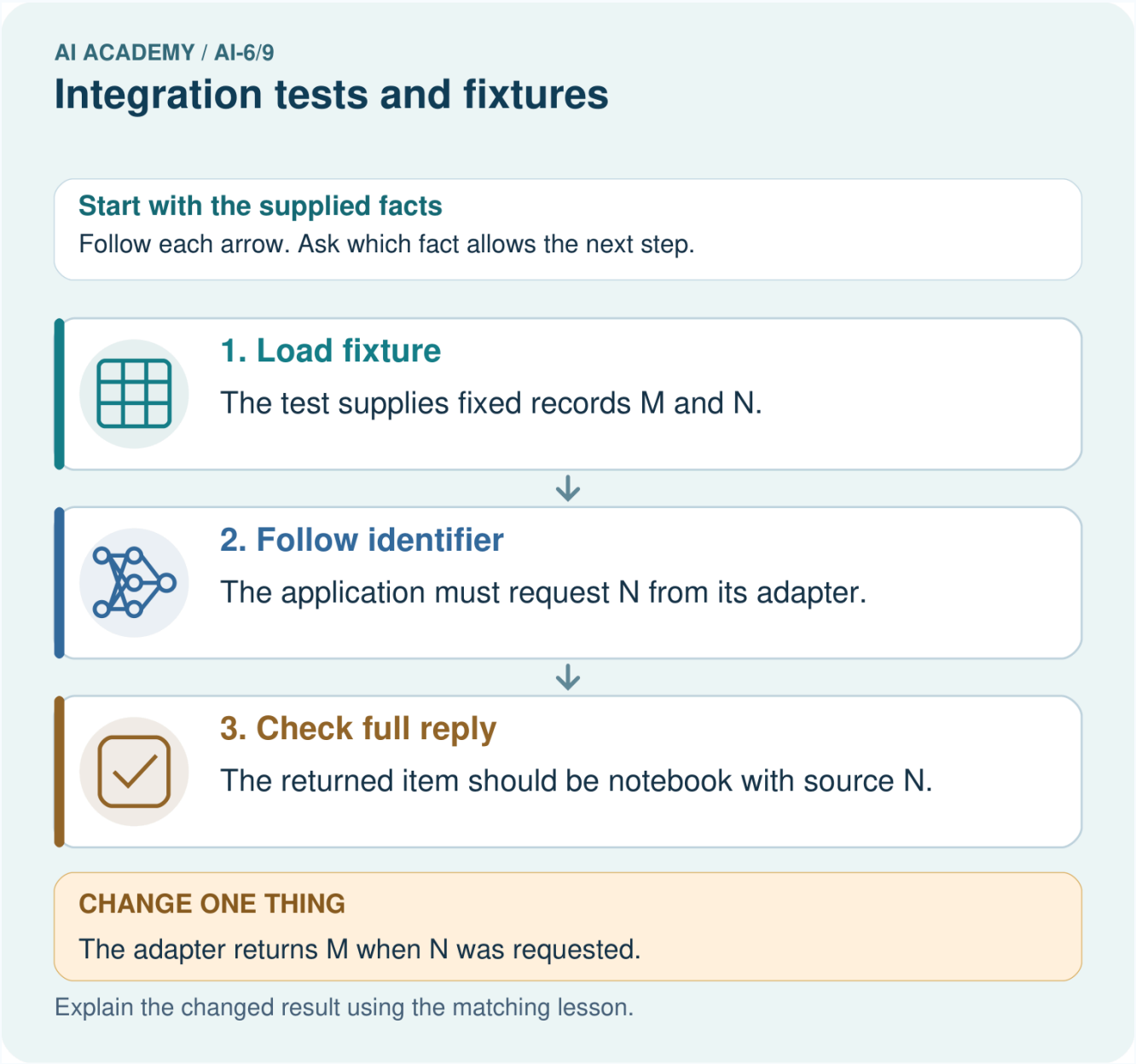
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Integration tests and fixtures

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The test exposes a hidden boundary failure despite the denied final response. Assert both response and **call trace**; deny before lookup.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Integration test**s and **fixture**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A mock verifies declared interactions, not real provider availability or all runtime limits.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Follow request N through the **fixture** and adapter. Which source and item should appear in the reply, and what precisely fails when the adapter returns M?

The test supplies fixed records M and N. The application must request N from its adapter. The returned item should be notebook with source N. Changed case: Fail the integration check: map is the wrong item.

The request now reaches **evidence** for another identifier.

Changed-case reasoning: Fail the integration check: map is the wrong item. The request now reaches **evidence** for another identifier.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main correct denial has empty restricted lookup trace; broken order reads T and fails despite same denial. Missing/failing S must review without draft stub. Catalogue requested N expects notebook/source N; M/map mismatch must fail/review. Independent storage-not-found case: evidence is absent, generation must not invent capacity, and final support validation must not label it supported.

Preserve failure trace, stop/review after missing evidence, and test no unsupported generation/publication.

Diagnose the current misconceptions

The worked result establishes every future case: A mock verifies declared interactions, not real provider availability or all runtime limits.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Valid student lookup gets not-found from storage. Generator invents a capacity and final response appears supported. Locate failures.

Teacher explanation: Not-found should route to explicit fallback, and semantic validation should reject invented capacity. Include this **fixture** and assert no supported result.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Integration test**s exercise connected components and failure propagation. **Fixture**s supply controlled **input** records and mocked service outcomes.

Trace request validation, session permission, lookup and **output** validation together. Include absent sources and failing dependencies; verify that denied requests never invoke restricted lookup.

W2 Trace the worked case

A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The test exposes a hidden boundary failure despite the denied final response. Assert both response and **call trace**; deny before lookup.

W3 Audit the limit

A report omits this qualification: A mock verifies declared interactions, not real provider availability or all runtime limits. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A mock verifies declared interactions, not real provider availability or all runtime limits.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A mock verifies declared interactions, not real provider availability or all runtime limits.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: The test exposes a hidden boundary failure despite the denied final response. Assert both response and **call trace**; deny before lookup.

Qualification: A mock verifies declared interactions, not real provider availability or all runtime limits.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Valid student lookup gets not-found from storage. Generator invents a capacity and final response appears supported. Locate failures.

Teacher explanation: Not-found should route to explicit fallback, and semantic validation should reject invented capacity. Include this **fixture** and assert no supported result.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Not-found should route to explicit fallback, and semantic validation should reject invented capacity. Include this **fixture** and assert no supported result. Require intermediate reasoning and the stated scope.

Fresh retry: Unauthenticated request returns denied after writing an audit record containing a teacher answer. Expected reasoning: Sensitive logging itself violates the boundary; test trace/log contents as well as status.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled **integration test**s and **fixture**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **integration test**s and **fixture**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Evaluation pipelines. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/10, ai-6/12, ai-6/15. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **integration test**s and **fixture**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The test exposes a hidden boundary failure despite the denied final response. Assert both response and **call trace**; deny before lookup.

Check every learner before discussion: Use the worked case for **Integration test**s and **fixture**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A mock verifies declared interactions, not real provider availability or all runtime limits.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The adapter returns M when N was requested. Require a **prediction** and a reason before comparing. Expected result: Fail the integration check: map is the wrong item.

Reason: The request now reaches **evidence** for another identifier.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

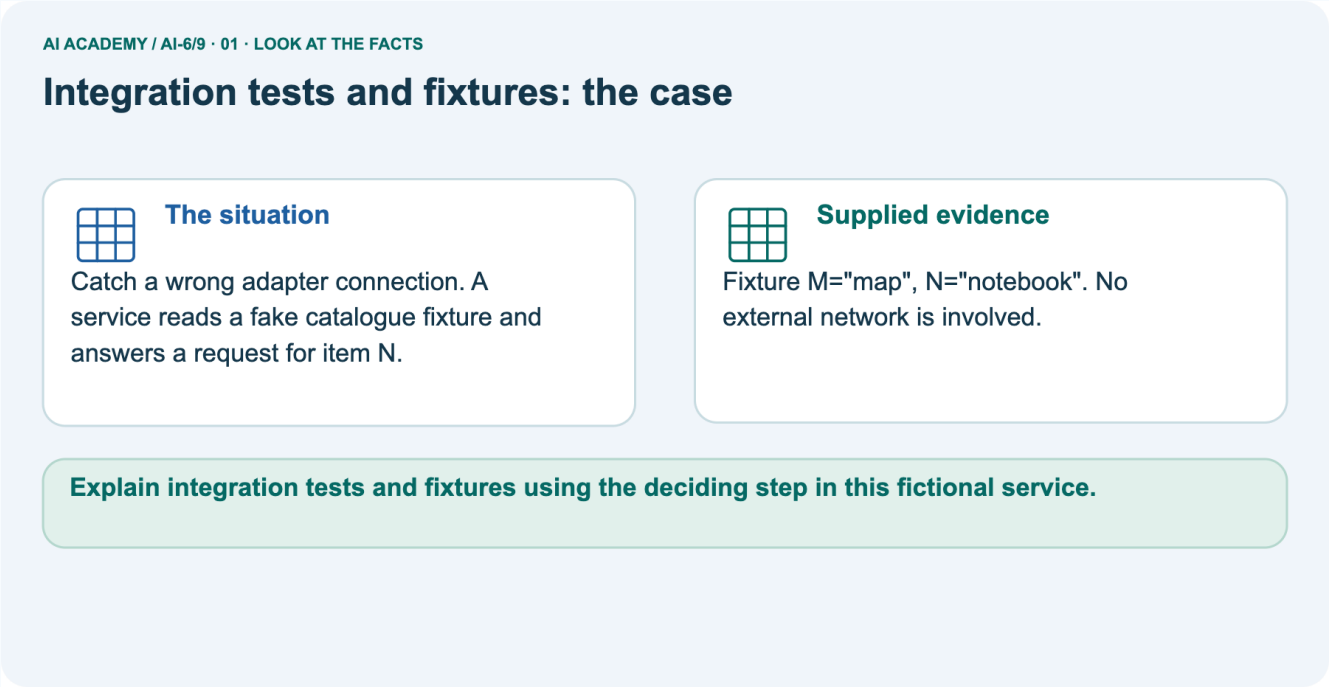
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Integration tests and fixtures: the case



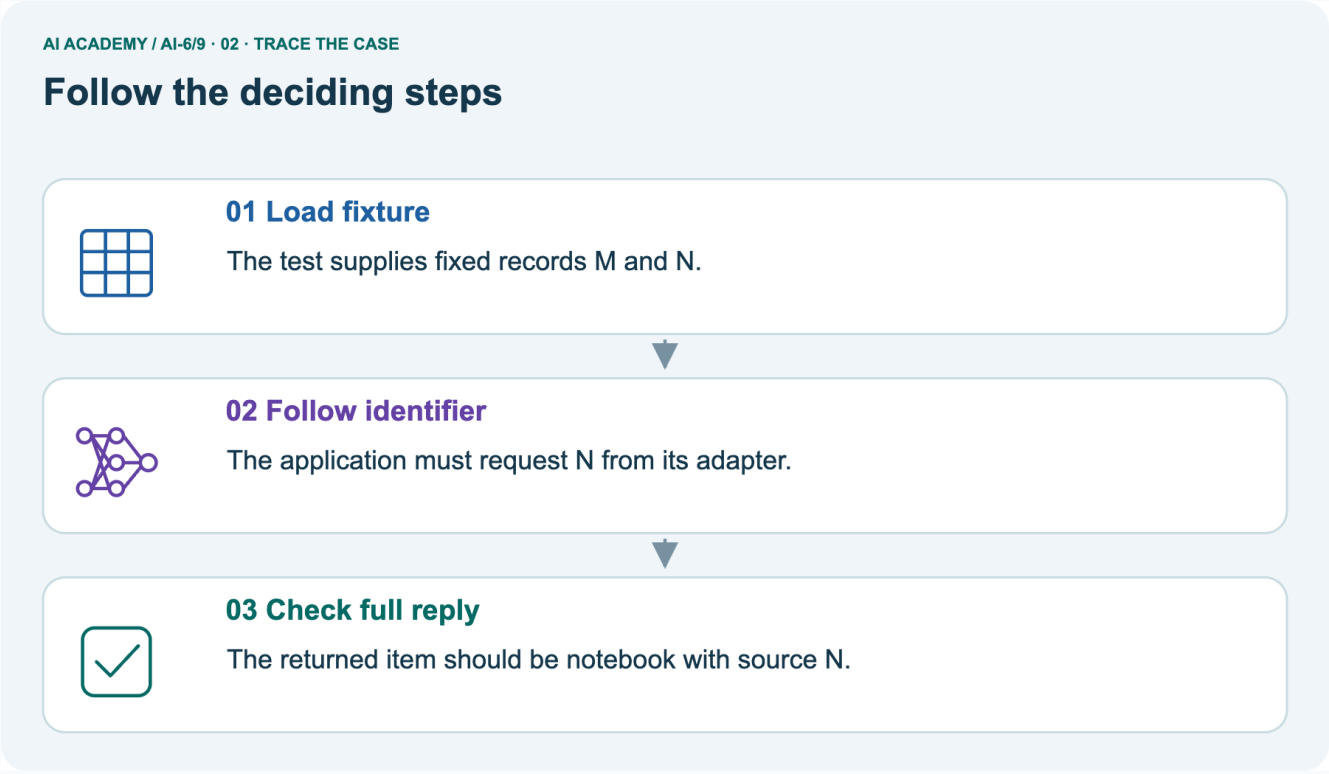
Follow the labelled picture with the matching explanation. Open the vector diagram

Catch a wrong adapter connection. A service reads a fake catalogue **fixture** and answers a request for item N. Supplied facts: **Fixture** M="map", N="notebook". No external network is involved.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



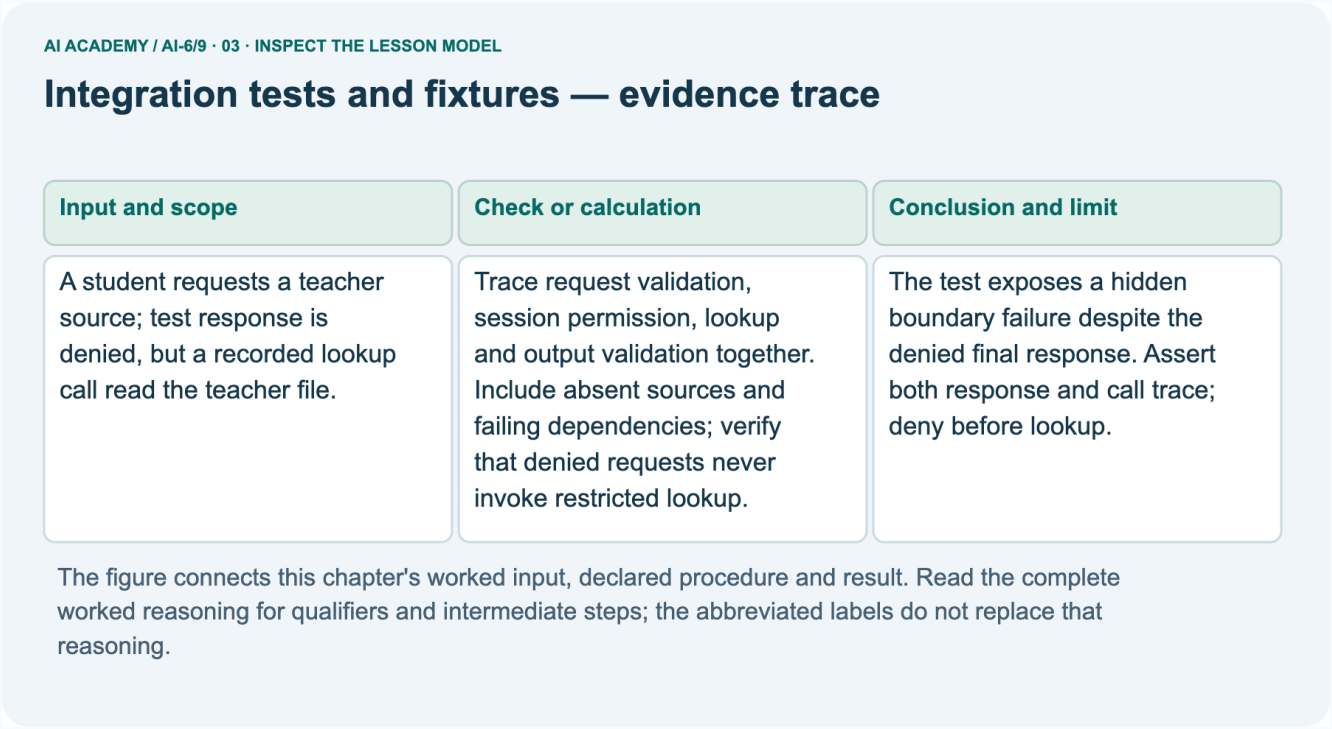
Follow the labelled picture with the matching explanation. Open the vector diagram

Load **fixture**: The test supplies fixed records M and N. → Follow identifier: The application must request N from its adapter. → Check full reply: The returned item should be notebook with source N.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Integration tests and fixtures — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Integration tests and **fixture**s — **evidence** trace.

A student requests a teacher source; test response is denied, but a recorded lookup call read the teacher file. | Trace request validation, session permission, lookup and **output** validation together.

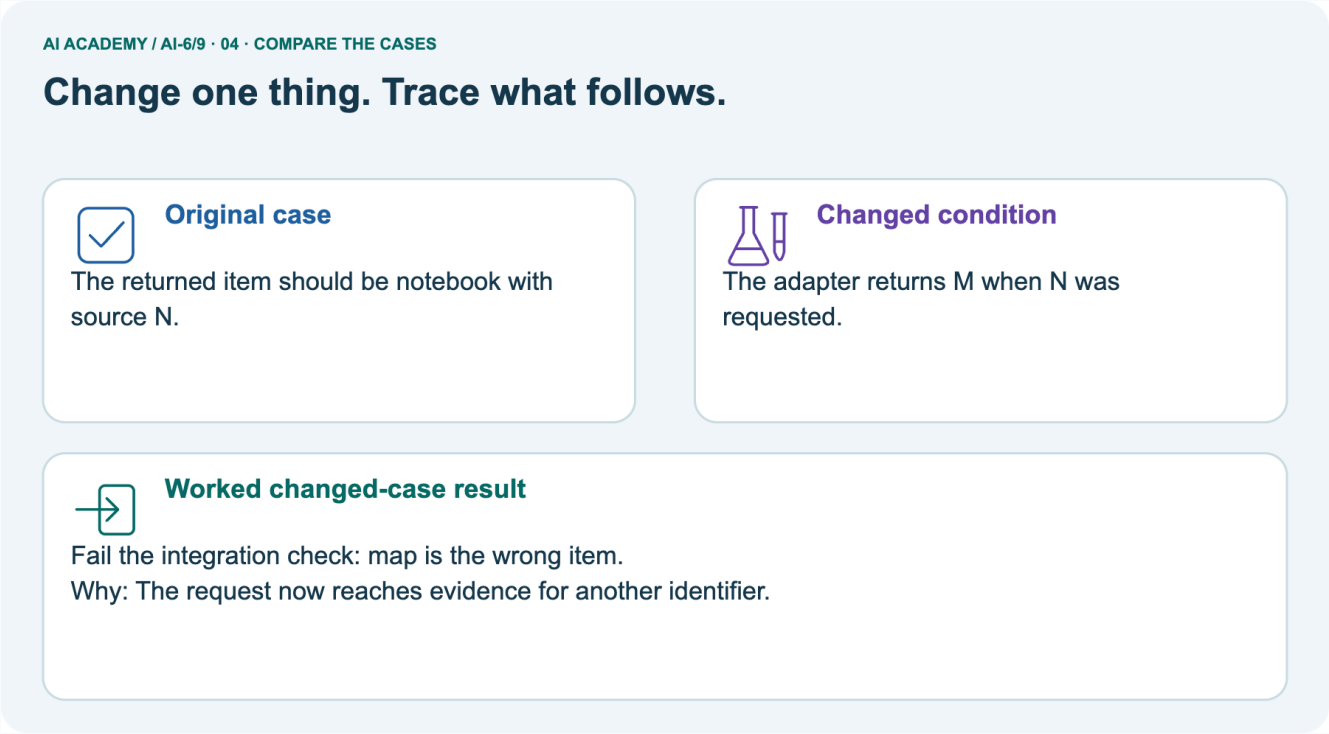
Include absent sources and failing dependencies; verify that denied requests never invoke restricted lookup. | The test exposes a hidden boundary failure despite the denied final response.

Assert both response and **call trace**; deny before lookup.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The returned item should be notebook with source N. Changed condition: The adapter returns M when N was requested. Expected result: Fail the integration check: map is the wrong item.

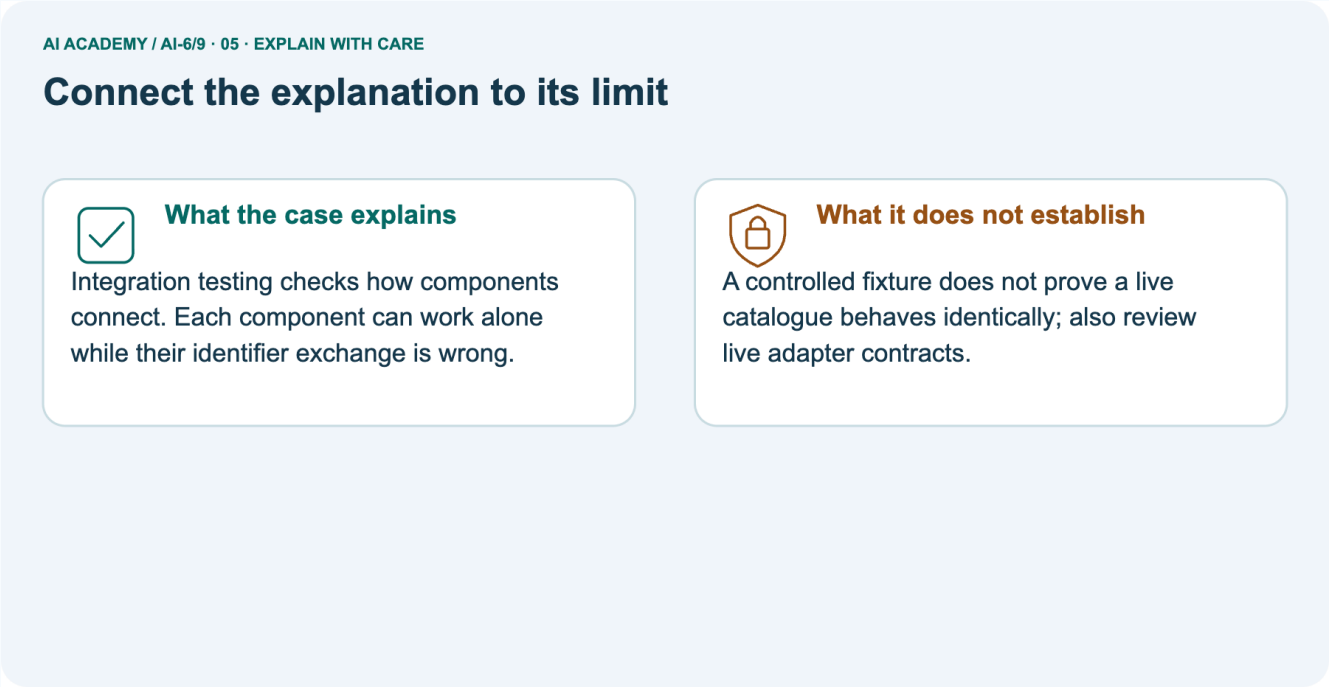
Reason: The request now reaches **evidence** for another identifier.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The request now reaches **evidence** for another identifier.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Integration testing checks how components connect. Each component can work alone while their identifier exchange is wrong.

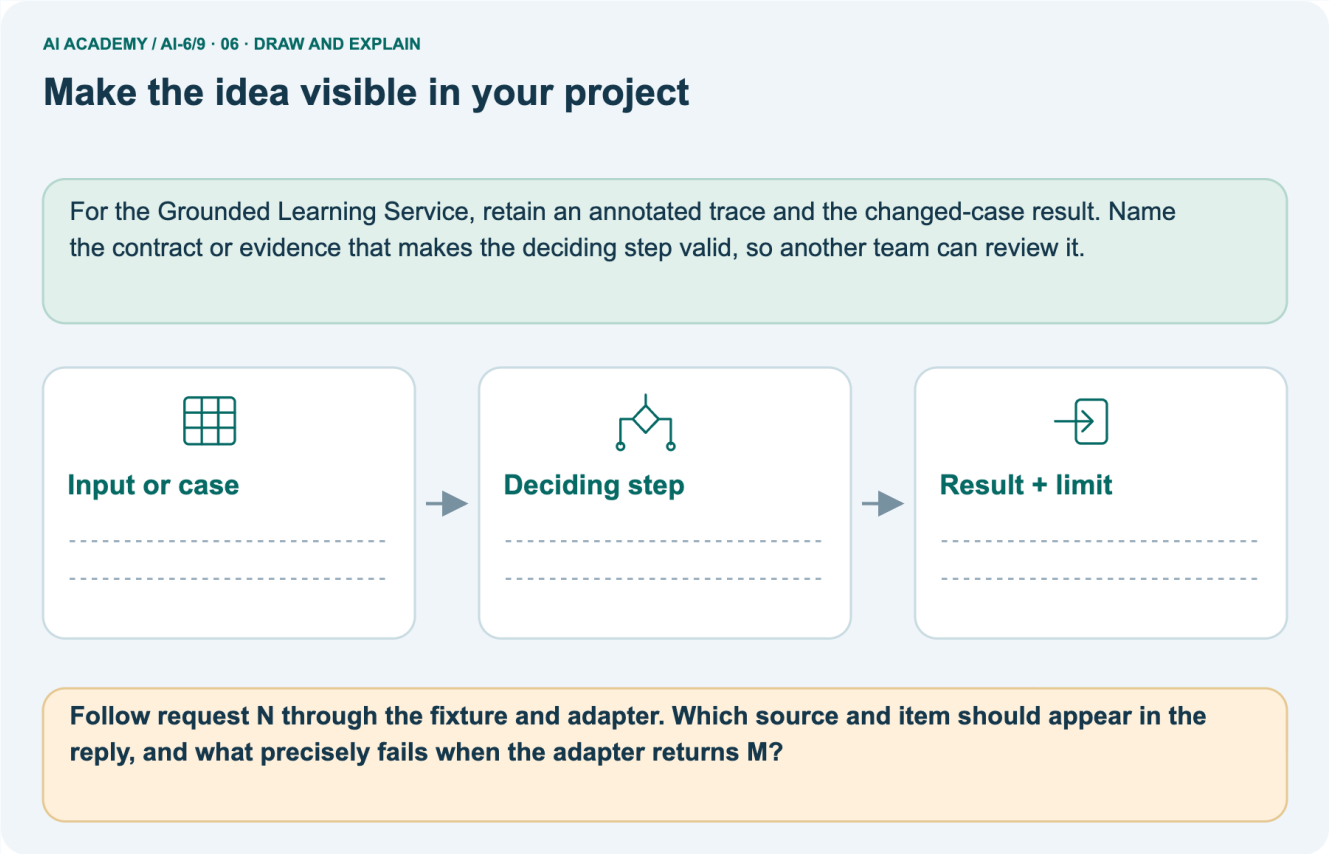
Limit: A controlled **fixture** does not prove a live catalogue behaves identically; also review live adapter contracts.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A controlled **fixture** does not prove a live catalogue behaves identically; also review live adapter contracts.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Follow request N through the **fixture** and adapter. Which source and item should appear in the reply, and what precisely fails when the adapter returns M?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The test supplies fixed records M and N. The application must request N from its adapter. The returned item should be notebook with source N.

Changed case: Fail the integration check: map is the wrong item. The request now reaches **evidence** for another identifier.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Integration tests and fixtures: the case

02. Follow the deciding steps

03. Integration tests and fixtures — evidence trace

04. Change one thing. Trace what follows.

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▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.